In [ ]:
```
ML Workflow:
Problem Statement
Data collection
```

In [ ]:
```
House prediction System Using ML & Streamlite

Business Problem:
A real -estate company wants to build a ml system
that can estimate the expected selling price of house
based on its characteristics:

Currently property agent may estimate prices
manually based on experience

This can lead to

-- Incosistance price
--slow property evalution
--subjective decisions
--difficult compairing many properties
--inaccurate initial pricing

Streamlite web app


Project Architecture:

House_price_Data

Data understanding

Data cleaning

EDA Analysis

Feature Engineering

Train Test Split

Data Processing

Numerical(Mean)               Categorical(Mode)
(Missing values)
```

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            Multiple LinearRegression(Try Diffrent regression algo)

Model Evalution

Best model selection

Save Model.pkl

Stramlite Application

User Enter Data
Price prediction
```

In [ ]:
```
Data Engineer
Data Analyst(Data Reading)

ML Engineer --ML Application pkl

DevOps Engineer  --->Deployment
```

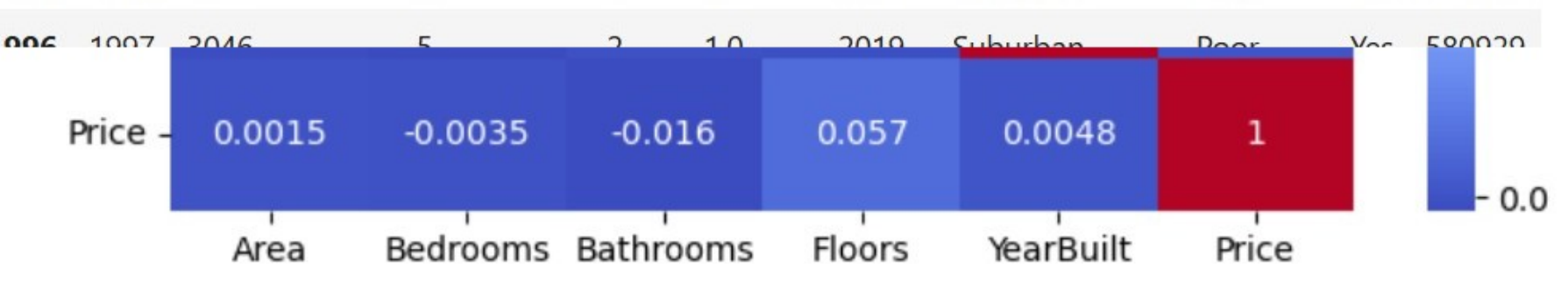
Data Collection

In [267…
```python
import pandas as pd
df=pd.read_csv(r"D:\Downloads\House Price Prediction Dataset (1).csv")
```

In [269…
```python
df
```

Out[269…

	Id	Area	Bedrooms	Bathrooms	Floors	YearBuilt	Location	Condition	Garage	Price
0	1	1360	5	4	3.0	1970	Downtown	Excellent	No	149919
1	2	4272	5	4	3.0	1958	Downtown	Excellent	No	424998
2	3	3592	2	2	3.0	1938	Downtown	Good	No	266746
3	4	966	4	2	2.0	1902	Suburban	Fair	Yes	244020
4	5	4926	1	4	2.0	1975	Downtown	Fair	Yes	636056
...	...	...	...	...	...	...	...	...	...	...
1995	1996	4994	5	4	3.0	1923	Suburban	Poor	No	295620

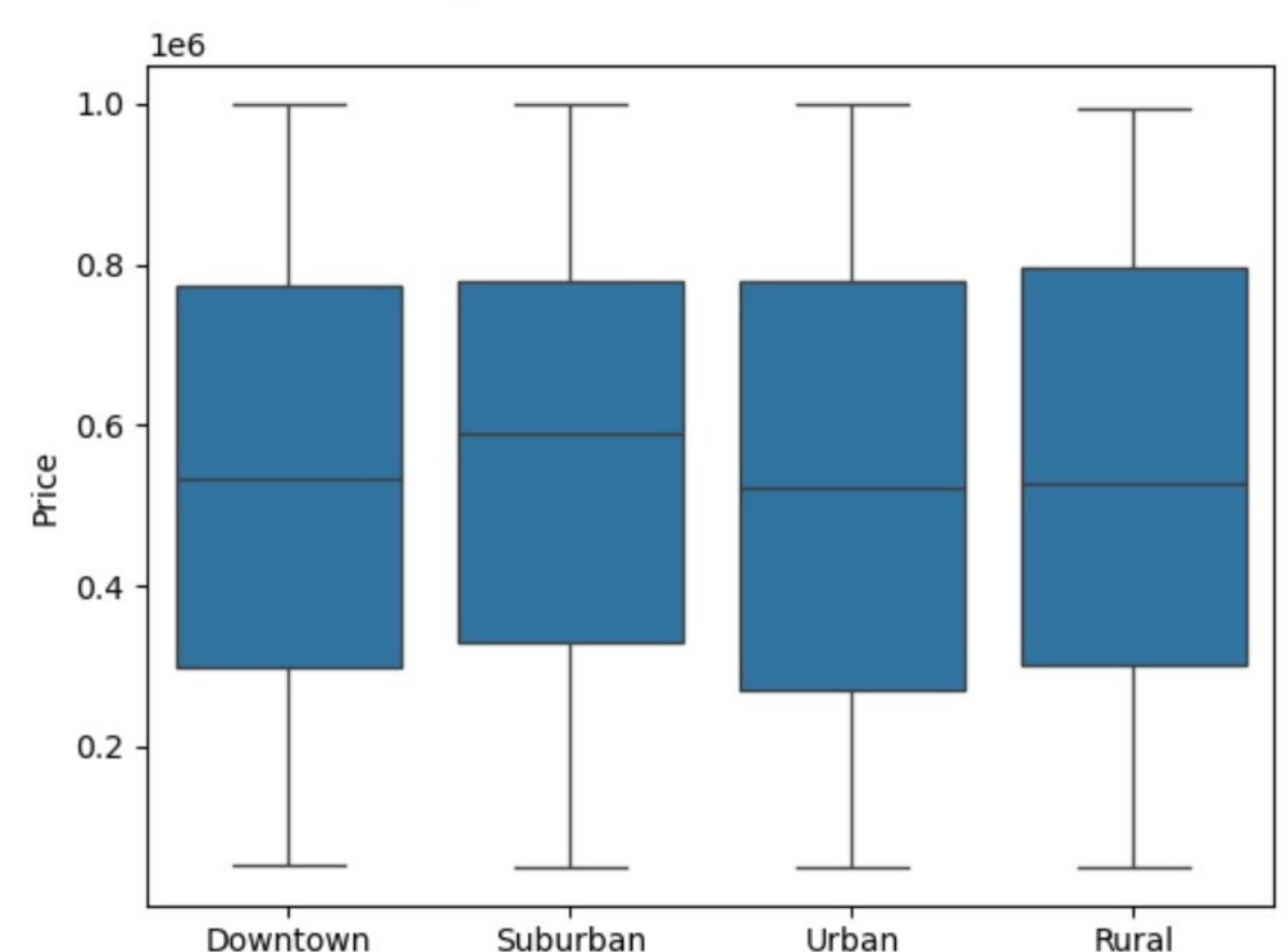


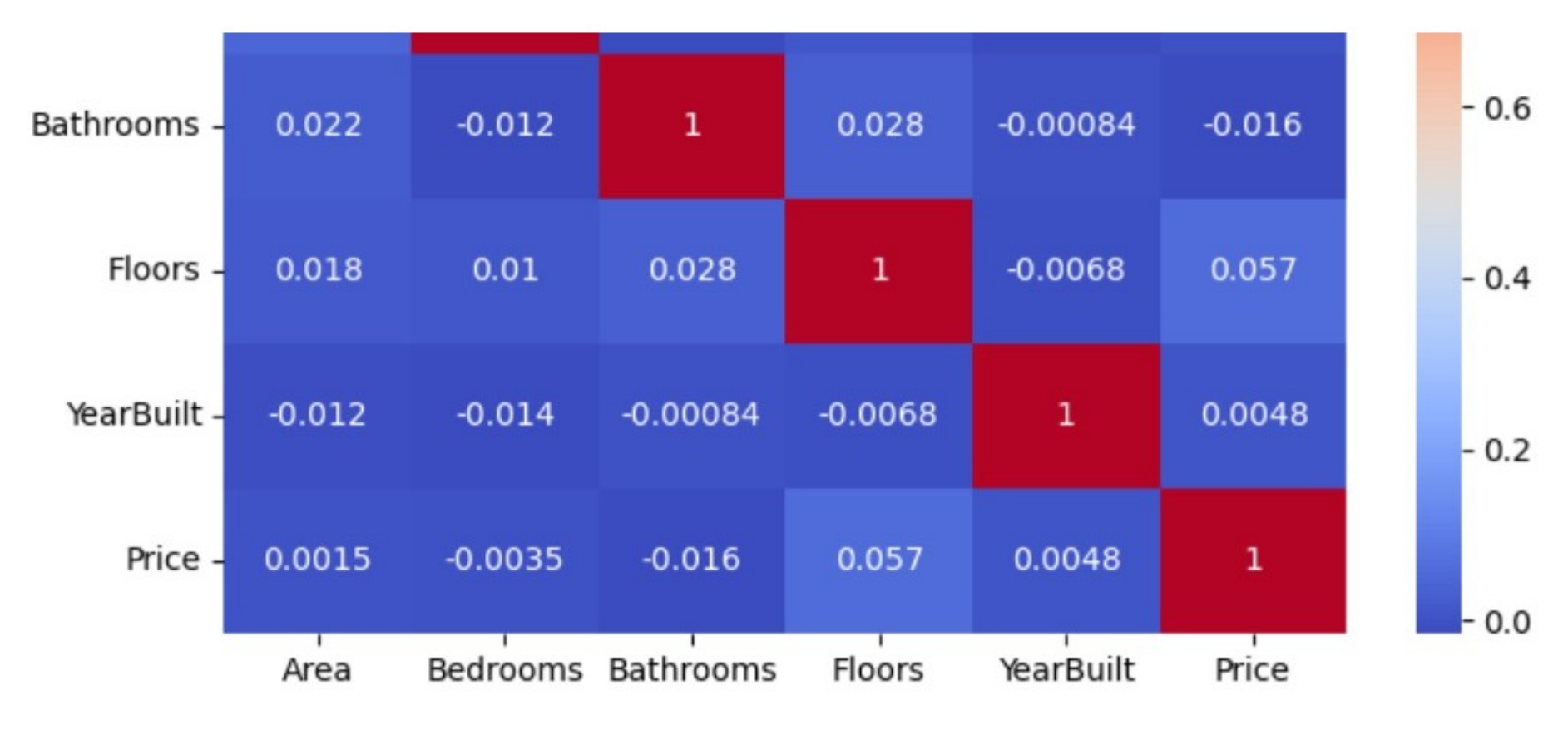
Categorical Analysis

Location Vs Price

In [312…
```python
sb.boxplot(data=df,
          x='Location',
          y='Price')
```

Out[312…
```
<Axes: xlabel='Location', ylabel='Price'>
```





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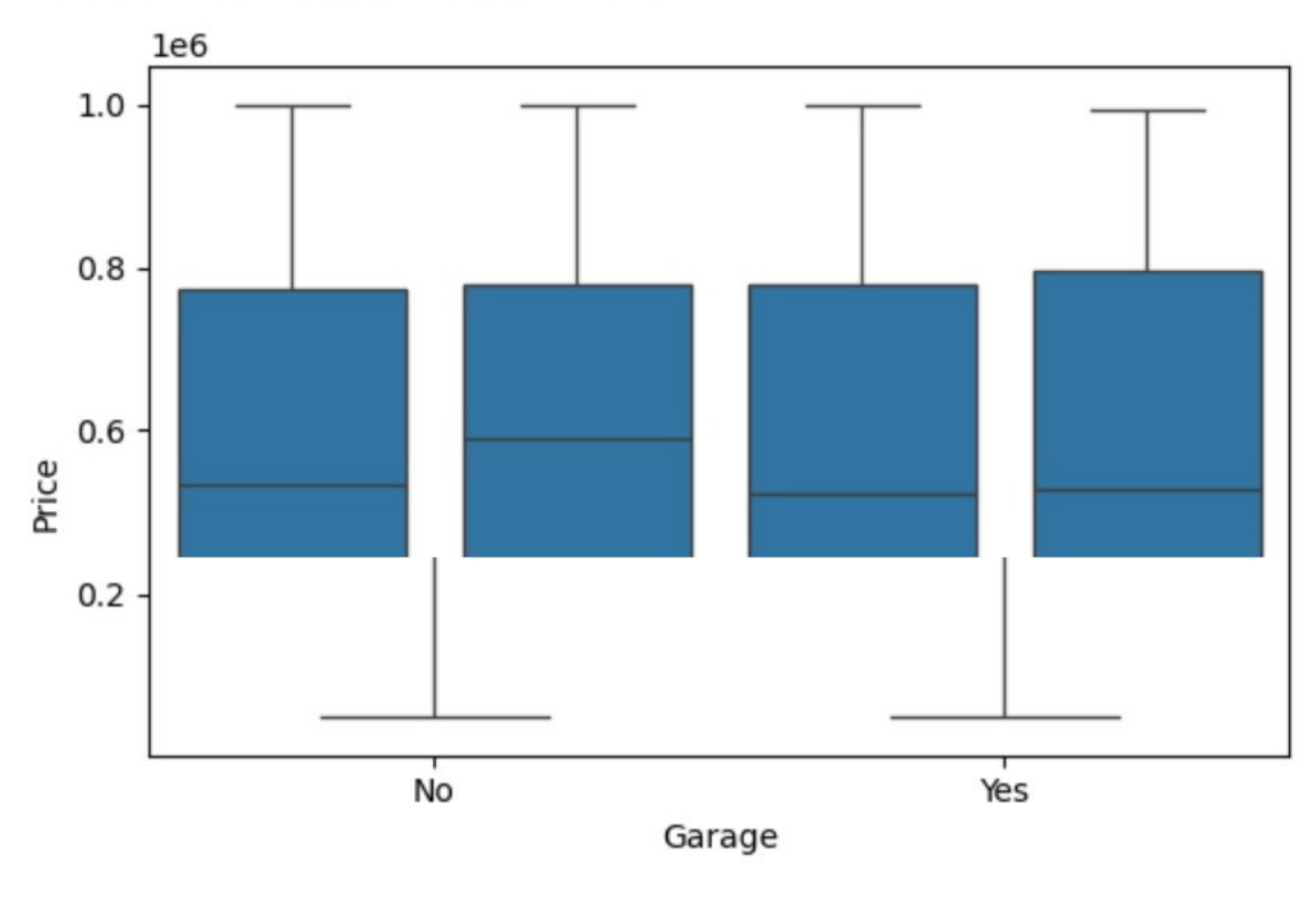

Define Target

In [321…
```python
Y=df[['Price']]
X=df.drop(labels=['Price'],axis=1)
```

In [323…
```python
Y
```

Out[323…

	Price
0	149919
1	424998
2	266746
3	244020
4	636056
...	...
1995	295620
1996	580929
1997	476925
1998	161119
1999	482525

2000 rows × 1 columns

In [325…
```python
X
```